

“**MELANCHOLY IS A HABIT, A SERIOUS AILMENT, A SETTLED HUMOUR, NOT ERRANT, BUT FIXED...**”

Robert Burton, English scholar, *The Anatomy of Melancholy*, 1621



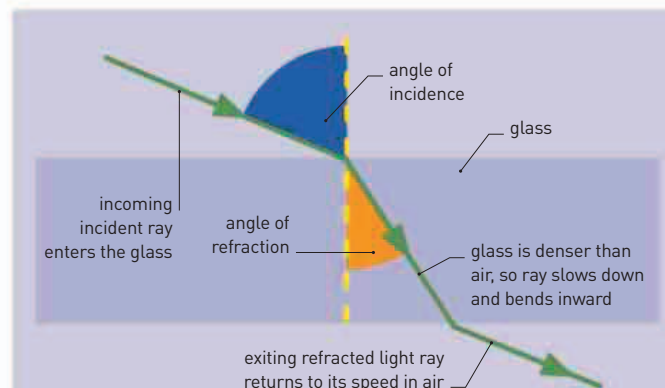
This engraving of the Greek philosopher Democritus is taken from the title page of Robert Burton's book on mental illness, *The Anatomy of Melancholy*.

William Bourne (c.1535–82) in 1578, and consisted of a wooden frame covered with leather and powered by oars. Drebbel subsequently built two **larger submarines, capable of carrying a number of passengers**, which were demonstrated on the Thames River in England. In tests, the final version of Drebbel's submarine managed to stay submerged for over three hours, suggesting that it had some means of providing oxygenated air for the occupants—although there are no records to explain how Drebbel could have achieved this. Despite the success of the submarine, the Royal Navy had no interest in using it.

In 1605, in the book *The Advancement of Learning*, English philosopher **Francis Bacon** had advocated **inductive reasoning for scientific investigation**. He wrote on the subject again in 1620, in a major treatise on logic called *Novum Organum Scientiarum* (*New Instrument of Science*). Bacon also advocated a process of reduction, which involved explaining the nature of things in terms of the relationships of their parts.

WITH HIS 1621 TREATISE ON THE CIRCLE, *Cyclometricus*, Dutchman Willibrord Snellius became the first European to publish the **law of refraction** describing the relationship between the angles of incidence and refraction of light passing through two different transparent substances, such as air and glass. Although known as **Snell's Law** (see panel, right), this principle had been mentioned by English mathematician Thomas Harriot about 20 years earlier, and was originally described by Persian mathematician Ibn Sahl in 984.

The best known book by English scholar **Robert Burton** (1577–1640), *The Anatomy of Melancholy*, appeared in 1621. It attempted to describe various forms of **mental disorder** and their symptoms, and suggested possible medical causes and remedies. Although the book was written in the style of a medical textbook, it was more a literary work than a scientific one. Nonetheless, it was a forerunner of later scientific studies of the psychology and psychiatry of mental disorders.

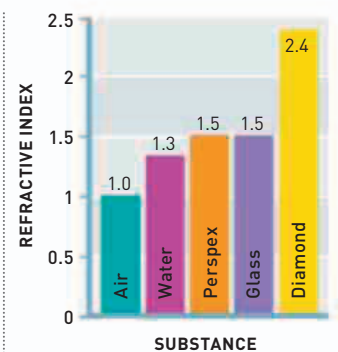


SNELL'S LAW

The law of refraction, or Snell's Law, concerns the relationship between the angle of incidence (the angle at which light approaches the surface of a transparent medium) and the angle of refraction (the angle the light takes as it changes speed through the medium). The relationship is constant for all angles of incidence and refraction, but varies from substance to substance.

Following from Napier's discovery of logarithms, English mathematician **Edmund Gunter** (1581–1626) devised **logarithmic scales** that could be engraved on a ruler to help seamen make calculations for navigation using a pair of compasses or dividers.

In 1622, English mathematician, **William Oughtred** (1574–1660), discovered that multiplication and division could be done by sliding two of Gunter's scales against each other and reading the result—the principle of the slide rule. Oughtred experimented



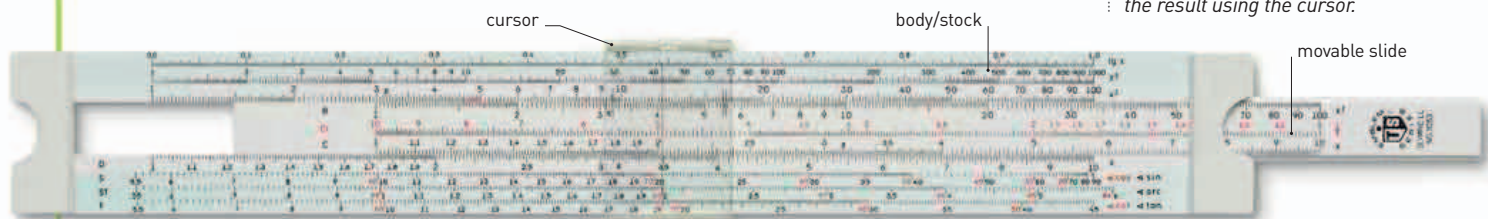
Refractive indices

The refractive index of a substance compares the speed of light when it passes through the substance to the speed of light in a vacuum.

with several designs for his slide rule, starting with a circular shape, but eventually settling on the familiar straight ruler with a sliding middle section, which remained in use until the invention of the pocket calculator some 300 years later.

Modern slide rule

Complex calculations could be done rapidly by lining up the different logarithmic scales inscribed on the rulers of the slide rule and reading the result using the cursor.



1620 Dutch inventor Cornelis Drebbel builds the first navigable submarine

1620 English philosopher Francis Bacon explains his method of **inductive reasoning** in *New Instrument of Science*

1621 Dutchman Willibrord Snellius rediscovers the Law of refraction

1621 English scholar Robert Burton publishes *The Anatomy of Melancholy*

1622 English mathematician William Oughtred invents the slide rule